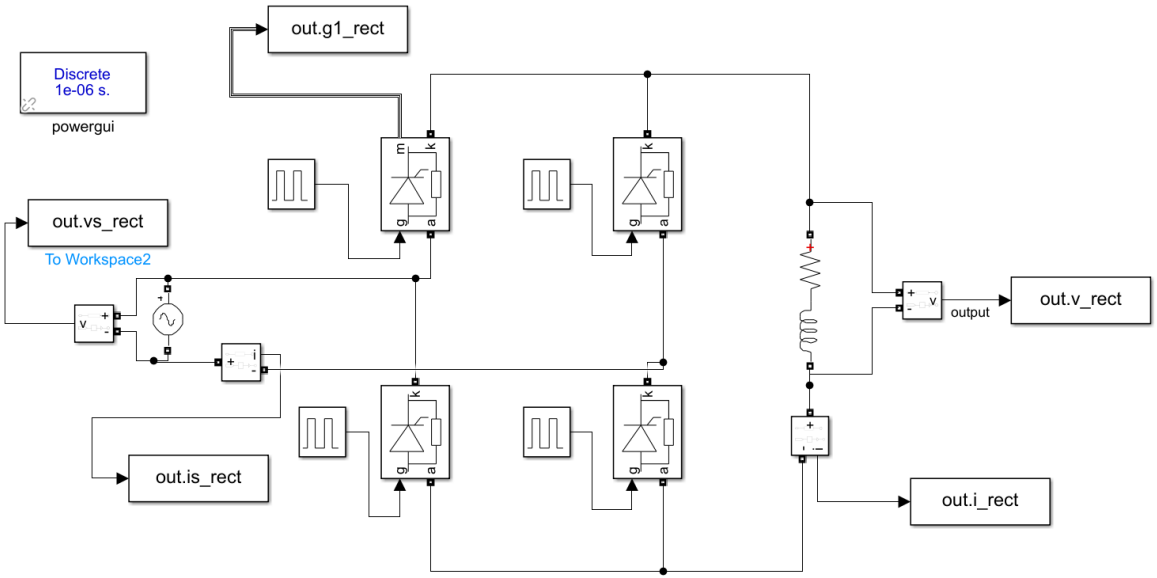


Figure 1: Single-phase full-wave rectifier model (RL load).

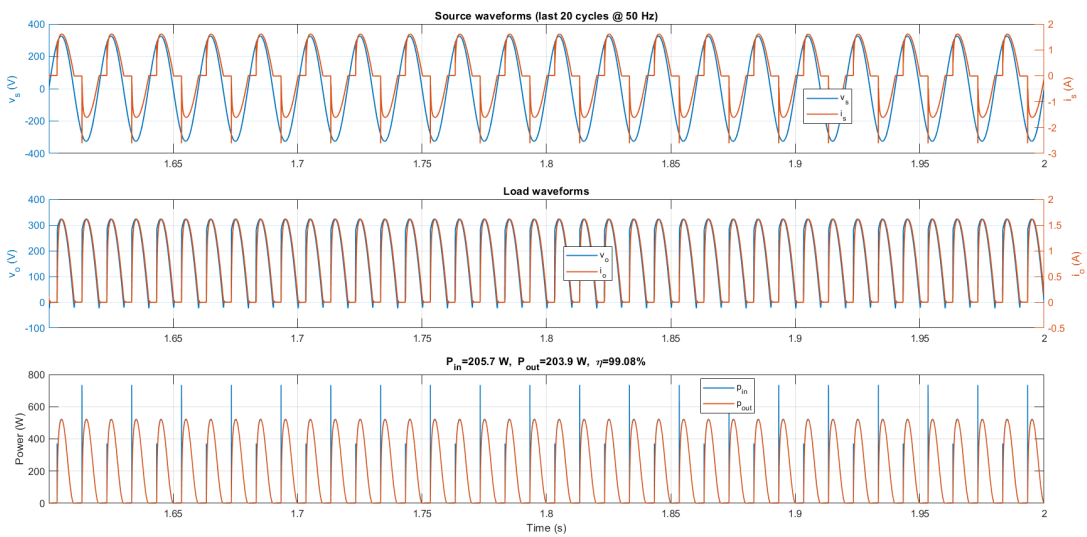


Figure 2: Representative source and load waveforms; bottom panel shows instantaneous P_{in} and P_{out}.

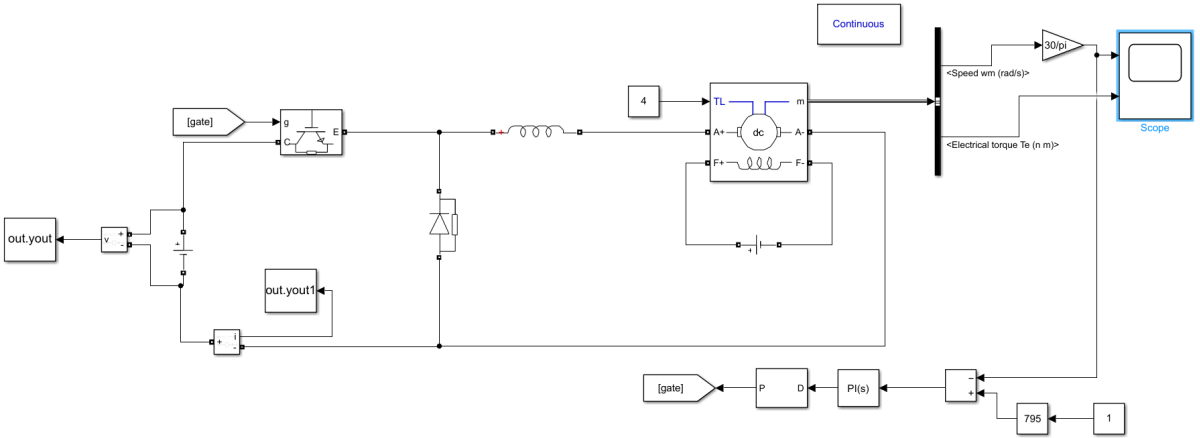


Figure 3: Buck chopper feeding a DC machine (speed loop PI).

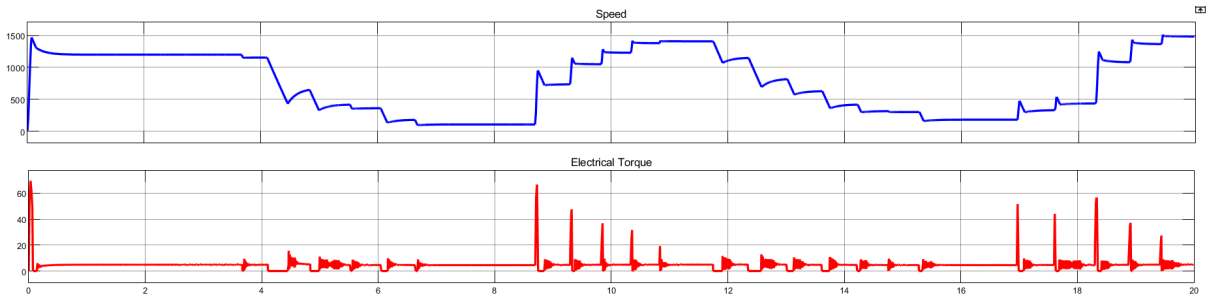


Figure 4: Measured speed (top) and electrical torque (bottom) of the DC drive for a stepped speed reference.

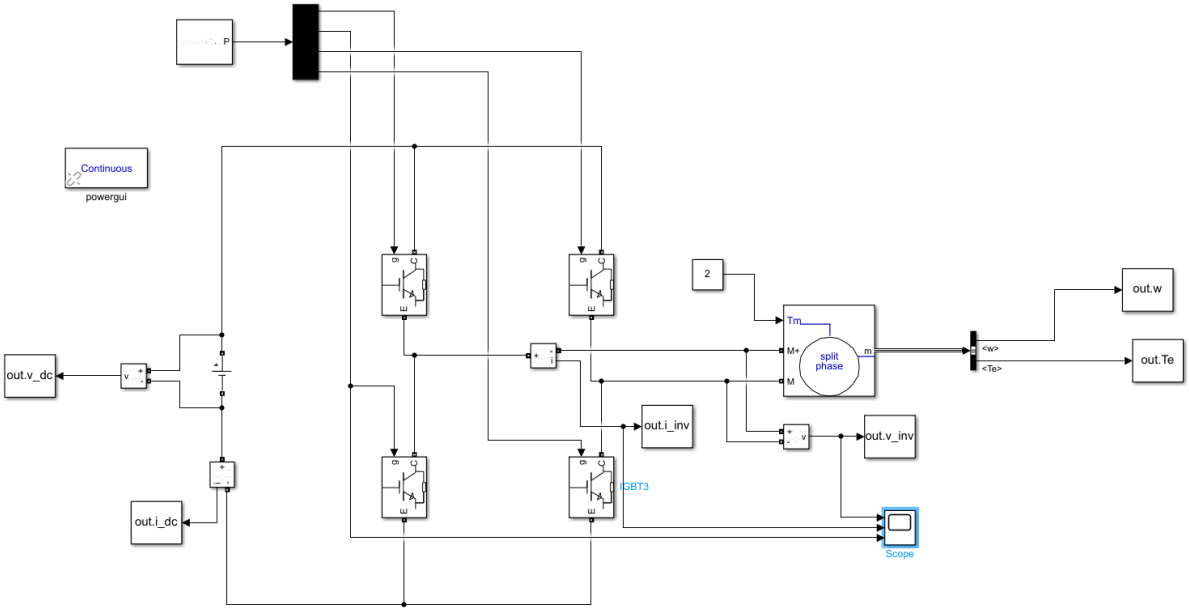


Figure 5: Single-phase VSI feeding a split-phase motor.

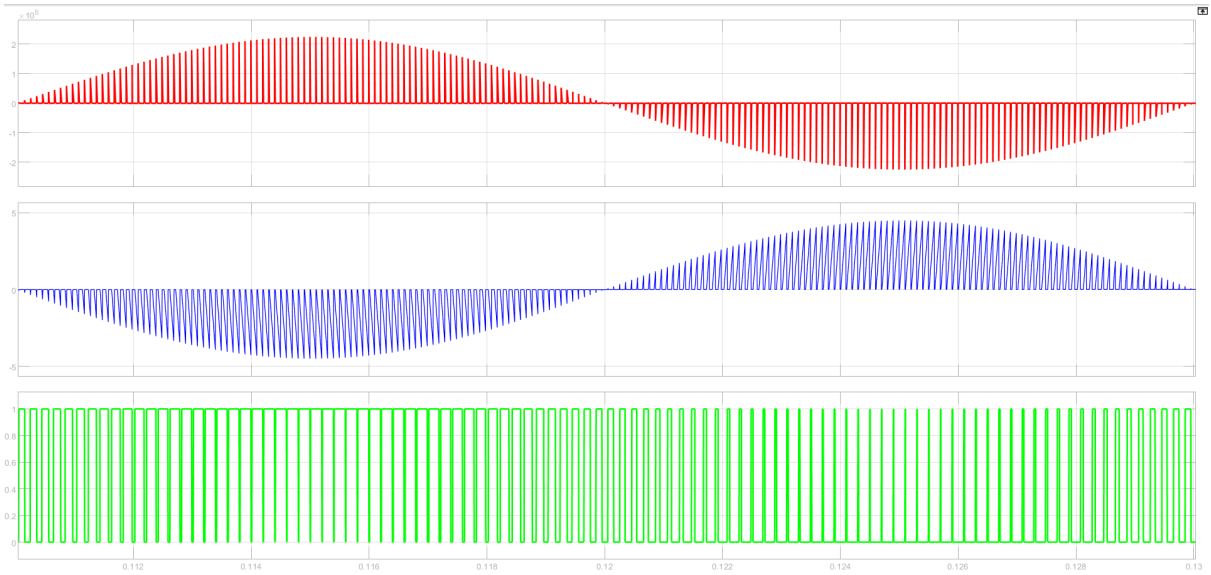


Figure 6: SPWM reference (bottom) and output voltage (top) with the corresponding output current (middle).